Survey Materials

This document contains the complete set of problems presented to evaluators in the study.

It is organized into a 10-problem calibration set and a 90-problem main study set; each problem is a geometry word problem to be rated.

Calibration Problems

1. A print shop has a triangular sticker design with sides measuring 9 cm, 12 cm, and 15 cm. To produce a smaller version for a keychain, they apply a dilation with a scale factor of 0.75 to the design. Find the length of the side on the smaller sticker that corresponds to the original 12 cm side.

2. A small triangular flag has two edges measuring 6 inches and 8 inches. A larger flag is made in the same shape, so the two flags are similar. On the larger flag, the edge that corresponds to the 6-inch edge measures 9 inches. Find the length of the edge on the larger flag that corresponds to the 8-inch edge.

3. Triangle ABC is similar to triangle DEF by AA similarity. In triangle ABC, AB = 9 and BC = 12. In triangle DEF, DE = 18. Find the length of DE.

4. Triangle PQR has PQ = 11 and QR = 13. Triangle XYZ has XY = 11, YZ = 13, and XZ = 15. Find the length of PR.

5. Triangle ABC has AB = 3, BC = 4, and AC = 9. Triangle ABC is dilated by a scale factor of 2 centered at the origin. Find the length of the image of AC.

6. Right triangle ABC has a right angle at B, with AB = 5, BC = 12, and AC = 14. Triangle DEF is similar to triangle ABC with a scale factor of 2. Find the length of DF.

7. In triangle ABC, angle A measures 52 degrees and angle C measures 63 degrees. Triangle ABC is similar to triangle DEF. Find the measure of angle B.

8. Consider a triangle, which we will call triangle ABC, that exists in the plane. This triangle ABC has three sides, and one of those sides, the side AB, has a length equal to 6 units. It is known that triangle ABC has been dilated, and the dilation that was applied to it used a scale factor whose value is 2.5, and the center of this dilation is a point that we call O. The resulting image is triangle A'B'C'. Determine, if you can, the length of the side A'B' of the image triangle A'B'C'.

9. Triangle ABC is dilated by a scale factor of 2. If AB = 5, find the length of A'B'.

10. In the figure, DE is parallel to BC. Given that AD = 4, DB = 6, and AE = 6, find the length of EC.

Main Study Problems

1. Let quadrilateral ABCD be a 4-sided figure with AB = 8, BC = 6, CD = 10, DA = 12. Let quadrilateral MNPQ be a 4-sided figure with NP = 3, PQ = 5. ABCD is dilated to form MNPQ with scale factor 0.5, centered at point P_center. Find the length of MN in MNPQ.

2. Let segment EF be a segment. Let segment GH be a segment with length 15. EF is dilated to form GH with scale factor 2.5, centered at point O. Find the length of EF.

3. Let triangle RST be a 3-sided figure with RS = 12, ST = 9, TR = 15. Let triangle UVW be a 3-sided figure with UV = 4, VW = 3, WU = 5. UVW is a scaled copy of RST. Find the length of RS in RST.

4. Let triangle ABC be a triangle with sides AB = 5, BC = 7, AC = 8. Let triangle A'B'C' be a triangle. ABC is dilated to form A'B'C' with scale factor 0.6, centered at point P. Find the length of A'B' in A'B'C'.

5. Let quadrilateral ABCD be a 4-sided figure with AB = 5, BC = 7, CD = 6, DA = 8. Let quadrilateral A'B'C'D' be a 4-sided figure. ABCD is dilated to form A'B'C'D' with scale factor 0.5, centered at point O. Find the length of B'C' in A'B'C'D'.

6. Let triangle PQR be a triangle. Let triangle P'Q'R' be a triangle. PQR is dilated to form P'Q'R' with scale factor unknown, centered at point C. Find the length of P'Q' in P'Q'R'.

7. Let triangle ABC be a triangle with sides AB = 14, BC = 16, AC = 10. Let triangle MNP be a triangle with sides NP = 8, MP = 5. Let segment MN be a midsegment segment. Let segment NP be a midsegment segment. Let segment MP be a midsegment segment. Let point M be a point on AB. Let point N be a point on BC. Let point P be a point on AC. ABC is dilated to form MNP with scale factor 0.5, centered at point A. Segment MN is parallel to AC. Segment NP is parallel to AB. Segment MP is parallel to BC. Find the length of MN.

8. Let triangle PQR be a triangle with sides PQ = 30. Let segment XY be a segment. Let segment PX be a segment with length 12. Let segment PY be a segment with length 9. Let point X be a point on PQ. Let point Y be a point on PR. Segments PX and XQ and PY and YR are proportional with ratio 0.6. By the triangle proportionality theorem, a line parallel to QR divides the other two sides proportionally into segments PX and PY. Find the length of XY.

9. Let triangle DEF be a triangle with sides EF = 20. Let triangle GHI be a triangle with sides GH = 7, HI = 10, GI = 8. Let segment GH be a midsegment segment. Let segment HI be a midsegment segment. Let segment GI be a midsegment segment. Let point G be a point on DE. Let point H be a point on EF. Let point I be a point on DF. DEF is dilated to form GHI with scale factor 0.5, centered at point E. Segment GH is parallel to DF. Segment HI is parallel to DE. Segment GI is parallel to EF. Find the length of DE in DEF.

10. Let triangle ABC be a triangle with sides AB = 6, BC = 8, AC = 10. Let triangle DEF be a triangle with sides DE = 9, EF = 12, DF = 15. Triangle ABC is similar to triangle DEF by SSS similarity with

scale factor 1.5. Find the length of EF in DEF.

11. Let quadrilateral PQRS be a 4-sided figure with PQ = 4, QR = 6, RS = 8, SP = 5. Let quadrilateral WXYZ be a 4-sided figure with WX = 6, XY = 9, YZ = 12, ZW = 7.5. Triangle PQRS is similar to triangle WXYZ by SSS similarity with scale factor 1.5. Find the length of RS in PQRS.

12. Let triangle MNO be a triangle with sides MN = 10, NO = 14, MO = 12, where ∠M = 45°, ∠N = 65°, ∠O = 70°. Let triangle PQR be a triangle with sides QR = 21, PR = 18, where ∠P = 45°, ∠Q = 65°, ∠R = 70°. Triangle MNO is similar to triangle PQR by AA similarity with scale factor 1.5 (congruent angle pairs: M ≅ P, N ≅ Q, O ≅ R). Find the length of PQ in PQR.

13. Let triangle ABC be a equilateral triangle with sides AB = 8, BC = 8, AC = 8, where ∠A = 60°, ∠B = 60°, ∠C = 60°. Let triangle DEF be a equilateral triangle with sides DE = 12, EF = 12, DF = 12, where ∠D = 60°, ∠E = 60°, ∠F = 60°. Triangle ABC is similar to triangle DEF by SSS similarity with scale factor 1.5. Find the length of AB in ABC.

14. Let point O be a point on c. Let point P be a point on d. Let circle c be a circle with radius 3. Let circle d be a circle with radius 7. c is dilated to form d with scale factor 7, centered at point O. Find the radius of circle d.

15. Let quadrilateral ABCD be a 4-sided figure with AB = 4, BC = 6, CD = 4, DA = 6, where ∠A = 90°, ∠B = 90°, ∠C = 90°, ∠D = 90°. Let quadrilateral PQRS be a 4-sided figure with PQ = 6, QR = 9, RS = 6, SP = 9, where ∠P = 90°, ∠Q = 90°, ∠R = 90°, ∠S = 90°. Triangle ABCD is similar to triangle PQRS by SAS similarity with scale factor 1.5 (congruent angle pairs: A ≅ P). Find the length of BC in ABCD.

16. Let triangle ABC be a triangle with sides AB = 10, where ∠A = 52°, ∠B = 68°. Let triangle DEF be a triangle with sides DE = 15, where ∠D = 52°, ∠F = 60°. The interior angles of triangle ABC (∠A, ∠B, ∠C) sum to 180°. The interior angles of triangle DEF (∠D, ∠E, ∠F) sum to 180°. Triangle ABC is similar to triangle DEF by AA similarity with scale factor 1.5 (congruent angle pairs: A ≅ D, B ≅ E). Find the measure of ∠E in DEF.

17. Let triangle PQR be a triangle with sides PQ = 12, QR = 9, where ∠P = 35°, ∠Q = 65°. Let triangle XYZ be a triangle with sides YZ = 18, where ∠Y = 80°, ∠Z = 35°. The interior angles of triangle PQR (∠P, ∠Q, ∠R) sum to 180°. The interior angles of triangle XYZ (∠X, ∠Y, ∠Z) sum to 180°. Find the measure of ∠X in XYZ.

18. Let triangle JKL be a triangle with sides JK = 5, where ∠J = 28°, ∠L = 112°. Let triangle MNO be a triangle with sides MN = 8, where ∠N = 40°, ∠O = 112°. The interior angles of triangle JKL (∠J, ∠K, ∠L) sum to 180°. The interior angles of triangle MNO (∠M, ∠N, ∠O) sum to 180°. Find the measure of ∠K in JKL.

19. Let triangle PQR be a triangle with sides PQ = 9, QR = 12, PR = 15. Let triangle XYZ be a triangle with sides XY = 6, YZ = 8. Triangle XYZ is similar to triangle PQR by SSS similarity with scale factor 1.5. Find the length of XZ in XYZ.

20. Let triangle MNO be a triangle with sides MN = 10, NO = 14, where ∠N = 48°. Let triangle RST be a triangle with sides RS = 15, ST = 21, where ∠S = 48°. Triangle MNO is similar to triangle RST by SAS similarity with scale factor 1.5 (congruent angle pairs: N ≅ S). Find the measure of ∠T in RST.

21. Let triangle JKL be a triangle with sides JK = 8, KL = 10, JL = 12, where ∠L = 42°. Let triangle DEF be a triangle with sides DE = 12, EF = 15, DF = 18. Triangle JKL is similar to triangle DEF by SSS similarity with scale factor 1.5 (congruent angle pairs: J ≅ D, K ≅ E, L ≅ F). Find the measure of ∠F in DEF.

22. Let triangle ABC be a right triangle with sides AB = 13, BC = 5. The interior angles of triangle ABC (∠A, ∠B, ∠C) sum to 180°. Find the length of AC in ABC.

23. Let triangle DEF be a right triangle with sides DE = 7, EF = 24. Let segment GE be a altitude segment perpendicular to DF. Altitude GE is drawn to hypotenuse DF, dividing it into segments DG and GF. Find the length of DF in DEF.

24. Let triangle PQR be a right triangle with sides PQ = 9, QR = 12, PR = 15. Let triangle PSR be a right triangle with sides PR = 15. Let segment QS be a altitude segment perpendicular to PR. Altitude QS is drawn to hypotenuse PR, dividing it into segments PS and SR. Altitude QS is the geometric mean of segments PS and SR. Find the length of QS.

25. Let triangle ABC be a triangle with sides AB = 12, BC = 9. The interior angles of triangle ABC (∠A, ∠B, ∠C) sum to 180°. Find the length of AC in ABC.

26. Let triangle PQR be a triangle with sides PQ = 7, QR = 24, PR = 25. The interior angles of triangle PQR (∠P, ∠Q, ∠R) sum to 180°. Determine whether triangle PQR is a right triangle.

27. Let triangle XYZ be a triangle with sides XY = 6, YZ = 8. The interior angles of triangle XYZ (∠X, ∠Y, ∠Z) sum to 180°. Find the length of XZ in XYZ.

28. Let triangle PQR be a right triangle with sides PQ = 9, QR = 12. Let triangle XYZ be a right triangle with sides XY = 6. The interior angles of triangle PQR (∠P, ∠Q, ∠R) sum to 180°. The interior angles of triangle XYZ (∠X, ∠Y, ∠Z) sum to 180°. Angles Q, Y are congruent. Triangle PQR is similar to triangle XYZ by AA similarity (congruent angle pairs: Q ≅ Y). Find the length of YZ in XYZ.

29. Let triangle DEF be a right triangle with sides DE = 5, EF = 12. Let triangle GHI be a right triangle with sides HI = 18. The interior angles of triangle DEF (∠D, ∠E, ∠F) sum to 180°. The interior angles of triangle GHI (∠G, ∠H, ∠I) sum to 180°. Angles E, H are congruent. Triangle DEF is similar to triangle GHI by AA similarity with scale factor 1.5 (congruent angle pairs: E ≅ H). Find the length of GH in GHI.

30. Let triangle JKL be a right triangle with sides JK = 8, KL = 6. Let triangle MNO be a right triangle with sides MN = 12. The interior angles of triangle JKL (∠J, ∠K, ∠L) sum to 180°. The interior angles of triangle MNO (∠M, ∠N, ∠O) sum to 180°. Angles L, O are congruent. Triangle JKL is similar to triangle MNO by AA similarity (congruent angle pairs: L ≅ O). Find the length of NO in MNO.

31. A graphic designer is creating a reduced version of a quadrilateral logo. The original logo is quadrilateral ABCD with side lengths AB = 8 cm, BC = 6 cm, CD = 10 cm, and DA = 12 cm. The designer uses a dilation with scale factor 0.5 and center O to shrink the logo to create quadrilateral MNPQ, where A → M, B → N, C → P, and D → Q.

What is the perimeter of the reduced logo MNPQ?

32. A graphic designer is creating a projected image of a company logo. The designer places a projector at point O and projects a small segment EF from the original logo onto a wall. The projector enlarges all distances from O by a scale factor of 2.5. The projected image segment GH on the wall measures 15 cm long.

To verify the setup, the designer measures the distance from the projector at O to point G (the image of E) and finds it to be 7.5 cm.

(a) Find the length of the original segment EF.

(b) Based on the definition of dilation, what is the distance from O to point E (the original location before enlargement)? Explain your reasoning in terms of how a dilation scales distances from the center.

33. An architect is designing a triangular sail for a new sailboat. The full-size sail RST has sides RS = 12 feet and ST = 9 feet. To preview the design, the architect creates a scale model UVW with a scale factor of 1/3 from triangle RST to triangle UVW.

The architect positions the scale model in a design software using a center of dilation P. The distance from P to point R on the full-size sail is 18 feet. Point U on the scale model is the image of point R under the dilation with center P and scale factor 1/3.

What is the distance from P to U?

34. A graphic designer is creating a scaled-down version of a triangular logo. The original logo is triangle ABC with sides AB = 5 cm, BC = 7 cm, and AC = 8 cm. The designer will dilate this triangle with a scale factor of 0.6, using point P as the center of dilation, to create the new logo triangle A'B'C'. Point P is located 10 cm from vertex A.

(a) What is the perimeter of the new scaled-down logo?

(b) How far is point P from vertex A' after the dilation?

35. A graphic designer is creating a pattern for a decorative tile. She has a quadrilateral design ABCD with sides AB = 5 cm, BC = 7 cm, CD = 6 cm, and DA = 8 cm. To create a smaller version of the design, she dilates the quadrilateral from a fixed center point O by a scale factor k, producing the image A'B'C'D'. After the dilation, the length of side B'C' is 3.5 cm.

(a) What is the scale factor k of this dilation?

(b) What is the length of side A'B' in the dilated quadrilateral?

(c) If the distance from O to vertex A is OA = 12 cm, what is the distance OA' from O to the image vertex A'?

36. A graphic designer is enlarging a triangular logo. The original triangle PQR has sides PQ = 8 cm and QR = 6 cm. The designer creates an enlarged version P'Q'R' by dilating triangle PQR from center point C with a scale factor of 5/2. What is the perimeter of the enlarged logo?

37. A triangular garden has sides of length 14 meters, 16 meters, and 10 meters. The gardener marks specific points on two sides of the garden such that these points divide those sides in the same ratio. Specifically, the points divide the 14-meter side and the 16-meter side in the ratio 2:1 from one vertex. The gardener plans to connect these two points with a narrow walking path.

If the 10-meter side is opposite the vertex from which the divisions are measured, what is the length of the walking path segment connecting the two marked points? Explain why this segment is parallel to the 10-meter side.

38. A designer is creating a triangular sail for a sailboat. The outer triangle PQR represents the full sail design, where PQ = 20 feet and PR = 15 feet, with QR = 25 feet.

To add a decorative reinforcement strip, the designer marks point X on side PQ and point Y on side PR such that the points divide the sides proportionally: PX = 12 feet and PY = 9 feet.

Use the triangle proportionality theorem to determine whether segment XY is parallel to QR. If it is parallel, find the length of XY in feet.

39. A triangular garden has vertices D, E, and F. Paved pathways connect the midpoints G, H, and I of sides DE, EF, and DF respectively, forming a smaller triangular region in the center of the garden.

The three pathways have lengths: GH = 5 meters, HI = 6 meters, and GI = 4 meters.

a) Explain why the pathway GH is parallel to side DF of the original garden.

b) What is the perimeter of the original triangle DEF?

40. A carpentry workshop is designing two triangular roof trusses. The first truss, ABC, has sides AB = 6 feet, BC = 8 feet, and AC = 10 feet. The second truss, DEF, is designed to be similar to truss ABC. On the workshop blueprint, truss DEF is shown in a different orientation: it has been rotated and is positioned differently from truss ABC. The sides of truss DEF are DE = 9 feet, EF = 12 feet, and DF = 15 feet.

Part (a): By examining the side lengths of both trusses, determine which vertices of DEF correspond to which vertices of ABC. Write a similarity statement relating triangle ABC to triangle DEF.

Part (b): What is the scale factor of the dilation that takes truss ABC to truss DEF?

Part (c): Describe a sequence of rigid motions and a dilation that would map truss ABC onto truss DEF. In your description, specify the center of the dilation and the scale factor. (Hint: You may use any rigid motions—such as a rotation, reflection, or translation—followed by a dilation, or in any order that works.)

Part (d): The carpenter needs to cut a beam for side EF of the second truss. How many feet of lumber should be cut for beam EF?

41. A quilter is designing two similar quilt panels. The smaller panel is quadrilateral PQRS with sides PQ = 4 cm, QR = 6 cm, RS = 8 cm, and SP = 5 cm. The larger panel is quadrilateral WXYZ with sides WX = 6 cm, XY = 9 cm, and ZW = 7.5 cm. The quadrilaterals are similar, and all pairs of corresponding angles are congruent.

(a) What is the scale factor of the dilation that takes quadrilateral PQRS onto quadrilateral WXYZ?

(b) Find the length of side YZ in quadrilateral WXYZ.

(c) Describe a sequence of rigid motions and a dilation that could take quadrilateral PQRS onto quadrilateral WXYZ. Be sure to specify the scale factor you use in your dilation.

42. Two landscape architects are designing triangular garden plots. The first garden plot is triangle MNO with sides MN = 10 feet, NO = 14 feet, and MO = 12 feet. A nearby school wants a larger version of this garden using the same shape but with NO = 21 feet. The larger garden is triangle PQR, where the sides correspond as follows: side MN corresponds to side PQ, side NO corresponds to side QR, and side MO corresponds to side PR.

(a) Write a similarity statement for the two triangles.

(b) What is the scale factor of the dilation that takes triangle MNO onto triangle PQR?

(c) Find the length of side PQ in the larger garden plot.

(d) Describe a sequence of rigid motions and a dilation that would map triangle MNO onto triangle PQR. Explain why your sequence works.

43. A landscape designer is creating two equilateral triangular garden tiles. The first tile, triangle ABC, has side length 8 feet. The second tile, triangle DEF, has side length 12 feet.

Part A: Prove that triangle ABC is similar to triangle DEF. In your proof, identify all pairs of

corresponding angles that are congruent, and explain why the side lengths of the two triangles are proportional (finding the scale factor).

Part B: Describe the sequence of transformations (rigid motions and dilations) that would map triangle ABC onto triangle DEF. How many dilations do you need, and why?

Part C: Contrast your approach with Tyler's failed proof that all rectangles are similar (from the lesson). Explain why one dilation is sufficient to prove that all equilateral triangles are similar, but two dilations would be needed for rectangles. What does this tell us about why not all rectangles are similar to each other?

44. A graphic designer is working with two circular logos. The original logo, circle c, has a center at point O and a radius of 3 inches. The client requests a scaled-up version, circle d, with center at point P and a radius of 7 inches. The two logos need to be positioned so that they appear as if one is an enlarged version of the other on a poster.

A colleague suggests the following sequence of transformations: First dilate circle c using center O and a scale factor of 7/3, then translate the image so its center coincides with point P.

(a) Describe an alternative sequence of transformations (translation followed by a dilation) that would take circle c exactly onto circle d, as shown in the lesson proof.

(b) Will both sequences result in circle c coinciding exactly with circle d? Explain why or why not, and justify your reasoning using what you know about rigid transformations and dilations.

45. A graphic designer is comparing two rectangular picture frames for a client project.

Frame A measures 4 inches wide and 6 inches tall. Frame B measures 6 inches wide and 9 inches tall.

The designer believes that because both frames are rectangles, they must be similar to each other. However, before ordering Frame B, she wants to verify this claim by checking whether the rectangles are truly similar using the definition of similarity: a figure is similar to another if there exists a sequence of rigid motions (translations, rotations, and/or reflections) followed by exactly one dilation that takes the first figure onto the second.

Are Frame A and Frame B similar? Justify your answer by explaining whether a sequence of rigid motions and a single dilation can map Frame A onto Frame B. If they are similar, state the required scale factor. If they are not similar, explain why no single dilation can work.

46. Two triangular support brackets are being designed for a warehouse shelving system. The brackets use the same angle measurements but are scaled to different sizes.

The smaller bracket, ABC, has side AB = 10 cm, side BC = 8 cm, angle A = 52°, and angle B = 68°.

The larger bracket, DEF, is a scaled version with side DE = 15 cm, angle D = 52°, and angle F = 60°.

First, use the Triangle Angle Sum Theorem to find angle C in triangle ABC and angle E in triangle DEF.

Then, determine whether the two triangles are similar using the Angle-Angle Triangle Similarity Theorem. If they are similar, find the length of side EF (round to the nearest tenth of a centimeter if necessary).

If the triangles are not similar, explain why they cannot be similar.

47. A welder in a steel fabrication shop is constructing two roof support brackets of different sizes. Both brackets must have the same shape to fit the architectural design, but they will be scaled to different proportions based on the building's requirements.

The first bracket (PQR) has angles ∠P = 35° and ∠Q = 65°, with the side PQ = 12 cm connecting these two angles.

The second bracket (XYZ) has angles ∠X = 80° and ∠Y = 65°, with the side YZ = 18 cm connecting the 65° angle to the third vertex.

(a) Use the Triangle Angle Sum Theorem to find the measure of ∠R in bracket PQR.

(b) Use the Triangle Angle Sum Theorem to find the measure of ∠Z in bracket XYZ.

(c) Explain why the two brackets are similar using the Angle-Angle Triangle Similarity Theorem by identifying two pairs of congruent corresponding angles.

(d) Based on the Angle-Angle Triangle Similarity Theorem, the vertex correspondence is P↔Z, Q↔Y, and R↔X. This means side PQ in triangle PQR corresponds to side ZY in triangle XYZ. Use these corresponding sides to find the scale factor from triangle PQR to triangle XYZ.

(e) In bracket PQR, the side QR = 9 cm. From the vertex correspondence, side QR corresponds to side YX in triangle XYZ. Using the scale factor, find the length of the corresponding side YX. How long should the welder cut the steel for the corresponding side in the larger bracket?

48. Two triangular support brackets are being manufactured for a construction project. The brackets must support equal load angles to function properly.

The first bracket is shaped like triangle JKL with ∠J = 28° and ∠L = 112°. One edge JK measures 5 cm.

The second bracket is shaped like triangle MNO with ∠N = 40° and ∠O = 112°. One edge MN measures 8 cm.

To ensure the brackets work as a matched pair, an engineer needs to verify they are geometrically similar and then determine the length of edge NO on the second bracket.

a) Find the measure of $\angle K$ in triangle JKL using the Triangle Angle Sum Theorem.

b) Find the measure of $\angle M$ in triangle MNO using the Triangle Angle Sum Theorem.

c) Which angles in triangle JKL correspond to which angles in triangle MNO? Explain how you can use the Angle-Angle (AA) Triangle Similarity Theorem to show that triangles JKL and MNO are similar.

d) Since the triangles are similar, identify which sides correspond to each other (JK corresponds to MN, and KL corresponds to NO) and find the scale factor from triangle JKL to triangle MNO using these sides.

e) Using the Law of Sines, calculate the length of side KL in triangle JKL: $KL = 5 \cdot \sin(28°)/\sin(40°)$. Round your answer to two decimal places.

f) A machinist needs to cut edge NO of the second bracket. Using the scale factor from part (d) and the length of KL from part (e), how long should edge NO be? Round your answer to two decimal places.

49. Two triangular truss braces are used in a construction project. The larger brace, PQR, has sides PQ = 10 cm, QR = 14 cm, and PR = 18 cm. A smaller brace, XYZ, is designed to be a scaled copy of the larger brace, with sides XY = 15 cm and YZ = 21 cm.

First, identify the scale factor by finding the ratio of corresponding sides for the two pairs of sides you know. Then use this scale factor to find the length of XZ.

Finally, verify that all three pairs of corresponding sides have the same ratio, confirming that triangle XYZ is similar to triangle PQR by the Side-Side-Side Similarity Theorem.

50. A sailmaker is designing triangular sail panels. The first panel has sides of length 10 inches and 14 inches, with an included angle of 48°. The second panel has sides of length 15 inches and 21 inches, with an included angle of 48°. A third panel has sides of length 15 inches and 21 inches, with an included angle of 65°.

(a) Are the first and second panels similar in shape? Explain your reasoning using the Side-Angle-Side Triangle Similarity Theorem.

(b) Are the first and third panels similar in shape? Explain your reasoning.

(c) For the first and second panels, identify the scale factor and use it to find the length of the third

side of the second panel if the third side of the first panel is 18 inches.

51. Two triangular garden plots are being designed for a community landscape project. The first plot, JKL, has side lengths JK = 8 feet, KL = 10 feet, and JL = 12 feet. The second plot, DEF, has side lengths DE = 12 feet, EF = 15 feet, and DF = 18 feet.

A landscape designer needs to know whether the two plots have the same shape so she can adapt her sprinkler layout from one plot to the other.

(a) Prove that triangles JKL and DEF are similar using the Side-Side-Side Triangle Similarity Theorem. In your proof, explain how a dilation followed by a sequence of rigid motions (or a congruence theorem) shows the triangles must be similar.

(b) Determine which angle in triangle DEF corresponds to angle L in triangle JKL. Explain your reasoning by identifying how the vertices correspond based on the side ratios.

(c) Use the Law of Cosines to find the measure of angle L in triangle JKL. Then state the measure of the corresponding angle in triangle DEF.

52. A construction crew is building a roof truss for a storage building. The truss is shaped like a right triangle with the peak at vertex A, where two rafters meet at a right angle. The horizontal tie beam at the base is BC, which is 13 meters long. The two rafters measure AB = 5 meters and AC = 12 meters. A vertical support beam (altitude) AD is drawn from the right-angle peak A perpendicular down to the tie beam BC, meeting it at point D. This altitude divides the tie beam into two segments: BD = x meters and DC = y meters.

In this configuration, the original large triangle ABC is similar to the two smaller triangles formed: triangle ABD and triangle ACD. Use this similarity relationship to answer the following:

(a) Write the proportion that relates corresponding sides of triangle ABC and triangle ABD. Use this proportion to derive the equation $AB^2 = BD \cdot BC$. Show your work.

(b) Similarly, write the proportion that relates corresponding sides of triangle ABC and triangle ACD. Use this proportion to derive the equation $AC^2 = CD \cdot BC$. Show your work.

(c) Using your derived equations from parts (a) and (b), compute the value of x (the length of BD) and the value of y (the length of DC).

(d) Finally, verify that the Pythagorean Theorem holds for this truss by showing that $AB^2 + AC^2 = BC^2$ using your computed values of x and y. What equation does this concrete calculation illustrate from the lesson's proof?

53. A rooftop support structure consists of a triangular frame DEF where the right angle is at E. The two legs of the support beam are DE = 7 meters and EF = 24 meters. To reinforce the structure, an engineer draws a support cable from vertex E perpendicular to the hypotenuse DF. Let G be the foot

of this perpendicular, so that G lies on DF and divides the hypotenuse into two segments, DG and GF.

(a) First, find the length of the hypotenuse DF using the Pythagorean Theorem.

(b) When an altitude is drawn to the hypotenuse of a right triangle, it creates three similar triangles. Identify which two of the original triangles (the large triangle DEF and the two smaller triangles formed by the altitude) are similar to each other in a way that relates the leg DE to the segment DG and the full hypotenuse DF. Using this similarity and the proportional relationship between corresponding sides, write an equation that expresses DE^2 in terms of DG and DF.

(c) Using your equation from part (b), find the length of segment DG.

(d) Find the length of segment GF and verify that DG + GF equals your answer to part (a).

54. A carpenter is building a roof truss support with a diagonal beam from one corner to another. The beam forms the hypotenuse of a right triangle with legs measuring 9 feet and 12 feet. The hypotenuse is 15 feet long.

The carpenter needs to install a vertical brace from the right-angle corner perpendicular to the diagonal beam, dividing the beam into two segments.

Let's call the right triangle PQR, where: - The two legs are PQ = 9 feet and QR = 12 feet - The hypotenuse is PR = 15 feet - A perpendicular brace from the right angle at Q meets the hypotenuse PR at point S - This creates segments PS and SR on the hypotenuse

Using the similar-triangle relationships from the lesson—specifically, that $a^2 = xc$ and $b^2 = yc$, where a and b are the legs, c is the hypotenuse, and x and y are the segments of the hypotenuse—find the length of the brace QS.

(Hint: Start by finding PS and SR using the similar-triangle proportions, then determine QS.)

55. A construction crew is building a triangular metal brace for a roof truss. The brace will have three sides measuring 9 feet, 12 feet, and 16 feet.

Before installation, the engineer needs to classify the triangle by its angles to ensure it meets safety standards. Use the converse of the Pythagorean Theorem to determine whether this triangular brace is acute, right, or obtuse. Explain your reasoning.

56. A carpenter is building shelves and needs to check whether the corner is truly square (forms a 90° angle). She measures three pieces that will form a triangle: two support braces measure 7 inches and 24 inches, and a diagonal brace measures 25 inches.

(a) Is the corner a true right angle? Use the converse of the Pythagorean Theorem to determine whether the triangle formed by these three measurements is a right triangle.

(b) Suppose the carpenter remeasures the diagonal and finds it is actually 26 inches instead of 25 inches. Without recalculating everything, would this new triangle with sides 7, 24, and 26 inches be acute, right, or obtuse? Explain your reasoning by comparing c^2 to $a^2 + b^2$.

57. A builder is framing a triangular roof support with two wooden beams. The two beams meet at point Y, with their other endpoints at points X and Z. One beam from Y to X has a length of 6 feet, and the other beam from Y to Z has a length of 8 feet. A third beam will connect points X and Z to complete the triangular frame.

Complete the inequality to describe the possible lengths for the third beam:

___ < x < ___

58. Two gardeners are designing similar right-triangular garden plots. In the larger plot PQR, the right angle is at Q, PQ = 9 meters, and QR = 12 meters. In the smaller plot XYZ, the right angle is at Y, and one leg XY = 6 meters corresponds to leg PQ in the larger plot. The angle at P in triangle PQR is congruent to the angle at X in triangle XYZ.

(a) Explain why triangle PQR is similar to triangle XYZ.

(b) What is the length of side YZ in the smaller garden plot?

(c) What is the length of the hypotenuse PR in the larger garden plot?

59. A metalworking company is creating two similar triangular support brackets for a structural reinforcement project. Both brackets are right triangles.

The first bracket, DEF, has a right angle at E, with DE = 5 inches and DF = 13 inches (the hypotenuse). The second bracket, GHI, is similar to DEF with a right angle at H, and side GI = 19.5 inches (the hypotenuse, corresponding to DF).

First, find the length of side EF in bracket DEF using the Pythagorean Theorem. Then, use the similarity relationship between the two triangles to find the length of side HI in bracket GHI, which corresponds to side EF.

What should the length of HI be?

60. Two triangular ramp designs for a skateboard park are being compared. Both ramps are right triangles.

The first ramp, represented by triangle JKL, has a right angle at K. The horizontal leg JK measures 8 meters, and the hypotenuse JL measures 10 meters.

The second ramp, represented by triangle MNO, is similar to the first ramp. It has a right angle at N, and the horizontal leg MN measures 12 meters.

The triangles are similar with the correspondence J ↔ M, K ↔ N, and L ↔ O.

What is the length of the vertical leg NO of the second ramp?

61. A graphic designer is preparing two versions of a logo: the original design ABCD and a smaller version MNPQ for use on business cards.

The original logo ABCD is a quadrilateral with sides AB = 8, BC = 6, CD = 10, and DA = 12. To create the business card version, the designer dilates the original from a fixed center point P_center using a scale factor of 0.5, producing the smaller quadrilateral MNPQ.

After the dilation, the designer measures two sides of the business card logo and finds that NP = 3 and PQ = 5. Before printing thousands of cards, she needs to verify her work by calculating the remaining side. What should the length of MN be in the dilated logo MNPQ?

62. A designer is creating a scaled drawing of a logo. She starts with an original segment EF and dilates it from a fixed center point O using a scale factor of 2.5, producing the enlarged segment GH. The dilated segment GH measures 15 units.

What was the length of the original segment EF?

63. A toy manufacturer is creating two different video game controllers. The original controller UVW has sides UV = 4 inches, VW = 3 inches, and WU = 5 inches. The company decides to make a larger version, controller RST, which is a scaled copy of UVW with a scale factor of 3.

The designer knows that ST = 9 inches and TR = 15 inches on the larger controller. To complete the design, she needs to find the length of the third side, RS, so she can order the correct amount of material for the casing.

What is the length of RS?

64. A graphic designer is working on a logo that features triangle ABC with sides AB = 5 cm, BC = 7 cm, and AC = 8 cm. To create a smaller version of the logo for use on business cards, she scales down the design from a fixed point P using a scale factor of 0.6, producing the reduced triangle A'B'C'.

When she prints the business card version, how long should the side A'B' be?

65. A designer is creating a scaled blueprint of a rectangular floor plan. The original floor plan ABCD has sides measuring AB = 5 meters, BC = 7 meters, CD = 6 meters, and DA = 8 meters. To fit the design on a smaller sheet of paper, she dilates the entire floor plan from a fixed reference point O

using a scale factor of 0.5, creating the reduced blueprint A'B'C'D'.

When she measures the corresponding side B'C' on the blueprint, what length should she find?

66. A surveyor is mapping out a triangular plot of land PQR with sides PQ = 4 meters, QR = 5 meters, and RP = 6 meters. To create a scaled version of this map for a development proposal, she dilates the original triangle from a fixed reference point C using a scale factor of 2, producing the enlarged triangle P'Q'R'.

When she prepares the proposal documents, she needs to know the length of side P'Q' on the enlarged map. What should that length be?

67. A triangular plot of land ABC has sides measuring AB = 14 meters, BC = 16 meters, and AC = 10 meters. A surveyor marks three points—M on side AB, N on side BC, and P on side AC—such that each point is positioned exactly halfway along its respective side from vertex A. These three midpoints form a smaller triangle MNP inside the original plot.

Because the surveyor placed each point at the midpoint of its side, the triangle MNP is actually a dilation of triangle ABC with a scale factor of 0.5, centered at point A. This means: - Segment MN is parallel to AC - Segment NP is parallel to AB - Segment MP is parallel to BC

The surveyor has already measured two sides of the smaller triangle: NP = 8 meters and MP = 5 meters.

To complete the survey, how long is the segment MN?

68. # Problem

A surveyor is mapping a triangular plot of land PQR with sides PQ = 30 meters and QR = 20 meters. She notices that a fence line runs from point X on side PQ to point Y on side PR, dividing those two sides into segments.

She measures: - PX = 12 meters - XQ = 18 meters - PY = 9 meters - YR = 6 meters

The surveyor suspects that the fence line XY might be parallel to the third side QR. If the fence really does divide the two sides proportionally (which the triangle proportionality theorem tells us means XY must be parallel to QR), how long is the fence line XY?

69. A triangular plot of land DEF has one side EF measuring 20 meters. A surveyor marks three points—G on side DE, H on side EF, and I on side DF—such that each of these points is positioned exactly halfway along its respective side. When she connects these three midpoints to form triangle GHI, she creates a smaller triangle inside the original one.

The surveyor measures the sides of the inner triangle GHI and finds they are GH = 7 meters, HI = 10 meters, and GI = 8 meters. She notices that: - Segment GH is parallel to side DF - Segment HI is

parallel to side DE - Segment GI is parallel to side EF

This inner triangle GHI is actually a dilation of the original triangle DEF, centered at vertex E with a scale factor of 0.5.

To complete her survey records, the surveyor needs to find the length of side DE in the original triangle DEF. What is this length?

70. A surveyor is mapping two triangular plots of land. The first plot, triangle ABC, has sides measuring AB = 6 meters, BC = 8 meters, and AC = 10 meters. The surveyor discovers that a second plot, triangle DEF, is similar to the first by SSS similarity, with a scale factor of 1.5. She has already measured two sides of the second plot: DE = 9 meters and DF = 15 meters.

To complete her map, she needs to find the length of the third side, EF. What should she record for the length of EF?

71. A designer is creating two similar decorative tile patterns. The first pattern, quadrilateral PQRS, has sides PQ = 4, QR = 6, and SP = 5. The second pattern, quadrilateral WXYZ, has sides WX = 6, XY = 9, YZ = 12, and ZW = 7.5.

The designer confirms that quadrilateral PQRS is similar to quadrilateral WXYZ using Side-Side-Side similarity, with a scale factor of 1.5 from PQRS to WXYZ.

To complete the first tile pattern, the designer needs to find the length of the remaining side RS in quadrilateral PQRS. What should this length be?

72. A surveyor is comparing two triangular plots of land. The original plot MNO has sides MN = 10 meters, NO = 14 meters, and MO = 12 meters. A developer wants to create a larger similar plot PQR for a new project.

The surveyor confirms that triangle MNO is similar to triangle PQR by AA similarity, with a scale factor of 1.5. This means each side of PQR is 1.5 times the corresponding side in MNO.

The surveyor has already measured two sides of the new plot: QR = 21 meters and PR = 18 meters. Before finalizing the land purchase, she needs to determine the third side of the plot.

What is the length of PQ?

73. A designer is comparing two equilateral triangle templates for a craft project. The first template, triangle ABC, has all sides measuring 8 units, with each angle measuring 60°. The second template, triangle DEF, has all sides measuring 12 units, with each angle also measuring 60°.

The designer notices that triangle DEF is a scaled-up version of triangle ABC—specifically, a dilation with scale factor 1.5. This means triangle ABC is similar to triangle DEF by the Side-Side-Side Similarity Theorem.

Since the designer knows that corresponding sides of similar triangles are proportional with the same scale factor, and she can see that DE = 12 and the scale factor from ABC to DEF is 1.5, what is the length of side AB in the original template triangle ABC?

74. Two optical lenses are being designed for a camera system. The first lens, circle c, has a radius of 3 millimeters. An engineer needs to create a second, larger lens, circle d, by enlarging the first design. She dilates circle c from a fixed point O on the circle using a scale factor of 7, which produces circle d. Point P lies on circle d.

What is the radius of the enlarged lens d?

75. I need to carefully analyze this input before proceeding.

Critical Issue Detected:

The function-translated problem contains a fundamental mathematical error that I cannot resolve by rewriting alone:

The problem states: - Quadrilateral ABCD with AB = 4, CD = 4, DA = 6, and all four angles = 90° - Quadrilateral PQRS with PQ = 6, QR = 9, RS = 6, SP = 9, and all four angles = 90° - Claims these are similar by SAS with scale factor 1.5

The Problem:

A quadrilateral with all four angles equal to 90° is a rectangle. For ABCD to be a rectangle, opposite sides must be equal. But we're given AB = 4, CD = 4, DA = 6, with no value for BC. This is incomplete and potentially contradictory (if it's a rectangle, BC must equal DA = 6).

For PQRS: PQ = 6, QR = 9, RS = 6, SP = 9. This *is* a valid rectangle (opposite sides equal).

If ABCD ~ PQRS with scale factor 1.5, then: - AB should correspond to PQ: 4 × 1.5 = 6 ✓ - DA should correspond to SP: 6 × 1.5 = 9 ✓

This means BC should correspond to QR: BC × 1.5 = 9, so BC = 6.

However, the problem asks me to "find the length of BC" — but the mathematical setup already determines it uniquely. The problem is solvable, but the input is malformed (it omits BC initially, then asks for it).

Additionally, the lesson context is about proving *categories* of shapes are similar (all circles, all equilateral triangles, etc.), not about applying similarity to a single pair of rectangles. The problem doesn't align with the lesson narrative.

My responsibility: I can only rewrite problems where the mathematics is correct and locked. The math here *is* correct (BC = 6 follows from the given similarity and scale factor), but the problem statement is incomplete/malformed in a way that suggests a transcription error.

I will proceed by assuming the intent is to find BC given the similarity relationship, and I will rewrite it naturally — but I must note that this problem does not align well with the lesson content provided.

A contractor is designing two rectangular storage rooms. The first room, ABCD, has dimensions AB = 4 meters and DA = 6 meters, with all corners at right angles. The second room, PQRS, is a larger version with dimensions PQ = 6 meters, QR = 9 meters, RS = 6 meters, and SP = 9 meters, also with all corners at right angles.

The contractor confirms that rectangle ABCD is similar to rectangle PQRS with a scale factor of 1.5, where angle A corresponds to angle P. Using this similarity relationship, what is the length of side BC in the smaller room ABCD?

76. A surveyor is examining two triangular plots of land. The first plot, triangle ABC, has a side AB = 10 meters, with angle A measuring 52° and angle B measuring 68°. The second plot, triangle DEF, has a side DE = 15 meters, with angle D measuring 52° and angle F measuring 60°.

The surveyor knows that the sum of all interior angles in any triangle equals 180°. She notices that angle A in the first plot matches angle D in the second plot (both 52°), and angle B in the first plot matches angle E in the second plot. This means the two triangular plots are similar by the Angle-Angle Similarity Theorem, with a scale factor of 1.5 relating corresponding sides.

To complete her survey report, the surveyor needs to find the measure of angle E in triangle DEF. What is $\angle E$?

77. A surveyor is measuring two triangular plots of land. The first plot, triangle PQR, has sides PQ = 12 meters and QR = 9 meters, with angles $\angle P = 35°$ and $\angle Q = 65°$. The second plot, triangle XYZ, has a side YZ = 18 meters, with angles $\angle Y = 80°$ and $\angle Z = 35°$.

The surveyor knows that the interior angles of any triangle must sum to 180°. Using this fact along with the given angle measures, she needs to find the measure of $\angle X$ in triangle XYZ so she can fully understand the shape of the second plot.

What is the measure of $\angle X$?

78. Let triangle JKL be a triangle with sides JK = 5, where $\angle J = 28°$, $\angle L = 112°$. Let triangle MNO be a triangle with sides MN = 8, where $\angle N = 40°$, $\angle O = 112°$. The interior angles of triangle JKL (its interior angles) sum to 180°. The interior angles of triangle MNO (its interior angles) sum to 180°. Triangle MNO is similar to triangle JKL by AA similarity (congruent angle pairs: $L \cong O$, $J \cong M$). Find the

measure of ∠K in JKL.

79. A surveyor is comparing two triangular plots of land. The first plot, triangle PQR, has sides measuring PQ = 9 meters, QR = 12 meters, and PR = 15 meters. The surveyor discovers a second, smaller plot, triangle XYZ, with two of its sides already measured: XY = 6 meters and YZ = 8 meters.

By checking the ratios of the corresponding sides, the surveyor confirms that triangle XYZ is similar to triangle PQR using the Side-Side-Side Similarity Theorem, with a scale factor of 1.5 (meaning each side of PQR is 1.5 times the length of the corresponding side in XYZ).

To complete the survey records, what is the length of the third side, XZ, in triangle XYZ?

80. A designer is comparing two triangular components for a mechanical system. The first triangle MNO has sides MN = 10 and NO = 14, with an angle of 48° at vertex N. The second triangle RST has sides RS = 15 and ST = 21, with an angle of 48° at vertex S.

The designer notices that the two triangles are similar by the Side-Angle-Side Similarity Theorem, with a scale factor of 1.5 (the angle at N is congruent to the angle at S).

Since the triangles are similar with this correspondence, what is the measure of angle T in triangle RST?

81. Let triangle JKL be a triangle with sides JK = 8, KL = 10, JL = 12, where ∠L = 42°. Let triangle DEF be a triangle with sides DE = 12, EF = 15, DF = 18. Triangle JKL is similar to triangle DEF by SSS similarity with scale factor 1.5 (congruent angle pairs: J ≅ D, K ≅ E, L ≅ F). Find the measure of ∠F in DEF.

82. Let triangle ABC be a right triangle with sides AB = 13, BC = 5. Find the length of AC in ABC.

83. Let triangle DEF be a right triangle with sides DE = 7, EF = 24. Let segment GE be a altitude segment perpendicular to DF. Altitude GE is drawn to hypotenuse DF, dividing it into segments DG and GF. Find the length of DF in DEF.

84. Let triangle PQR be a right triangle with sides PQ = 9, QR = 12, PR = 15. Let triangle PSR be a right triangle with sides PR = 15. Let segment QS be a altitude segment perpendicular to PR. Altitude QS is drawn to hypotenuse PR, dividing it into segments PS and SR. Altitude QS is the geometric mean of segments PS and SR. Find the length of QS.

85. Let triangle ABC be a right triangle with sides AB = 12, BC = 9. Find the length of AC in ABC.

86. Triangle PQR has side lengths PQ = 7, QR = 24, and PR = 25. Using only these measurements, determine whether triangle PQR is a right triangle, and explain your reasoning.

87. Let triangle XYZ be a right triangle with sides XY = 6, YZ = 8. Find the length of XZ in XYZ.

88. A contractor is designing two similar roof trusses for a pair of buildings. The first truss, triangle PQR, is a right triangle with legs PQ = 9 feet and QR = 12 feet, meeting at the right angle Q. The second truss, triangle XYZ, is also a right triangle with a right angle at Y, and one of its legs measures XY = 6 feet.

The contractor knows that the two trusses are similar by AA similarity — specifically, the angles at Q and Y are congruent (both right angles), and the angles at P and X are also congruent. Because the trusses have the same shape, the contractor can use the proportional relationships between corresponding sides to find the unknown dimensions.

What is the length of side YZ in the smaller truss XYZ?

89. Let triangle DEF be a right triangle with sides DE = 5, EF = 12. Let triangle GHI be a right triangle with sides HI = 18. Triangle DEF is similar to triangle GHI by AA similarity with scale factor 1.5 (congruent angle pairs: E ≅ H). Find the length of GH in GHI.

90. A surveyor is mapping two similar right triangles formed by sight lines across a ravine. Triangle JKL has a right angle at K, with sides JK = 8 and KL = 6. Triangle MNO is similar to triangle JKL by AA similarity, with the angle at O matching the angle at L. The surveyor knows that one side of triangle MNO measures MN = 12.

Using the proportional relationships between the similar triangles, find the length of side NO in triangle MNO.